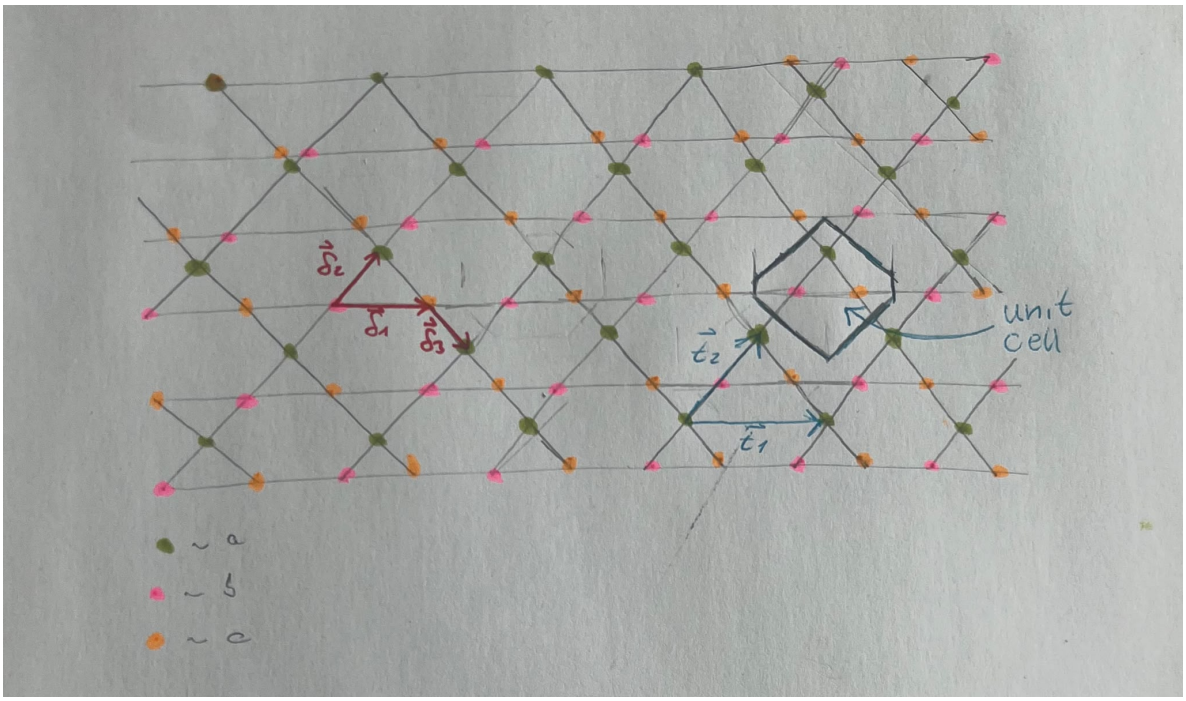
0.1 0-flux State

0.1.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise ;)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \tag{1}$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \tag{2}$$

The tight binding hamiltonain is

$$H = t\sum_{i\sigma}\left(a^\dagger_{i\sigma}c_{i\sigma} + c^\dagger_{i\sigma}b_{i\sigma} + b^\dagger_{i\sigma}a_{i\sigma} + c^\dagger_{i\sigma}b_{i+t_1;\sigma} + c^\dagger_{i\sigma}a_{i+t_1-t_2;\sigma} + a^\dagger_{i\sigma}b_{i+t_2;\sigma} + \text{h.c.}\right) \tag{3}$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.1.2 Reciprocal Space

The recipricoal lattice vectors are

$$\vec{b}_1 = \pi\begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi\begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \tag{4}$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \sqrt{\frac{3}{N}} \sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \tag{5}$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} & c^\dagger_{q\sigma} \end{pmatrix} \begin{pmatrix} 0 & \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & \cos\left(\vec{q}\cdot\vec{\delta}_3\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & 0 & \cos\left(\vec{q}\cdot\vec{\delta}_1\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_3\right) & \cos\left(\vec{q}\cdot\vec{\delta}_1\right) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \tag{6}$$

0.1.3 dispersion and band touching points

The three eigenvalues are

$$\varepsilon_1(\vec{q}) = t\left(1 - \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{7}$$

$$\varepsilon_2(\vec{q}) = t\left(1 + \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{8}$$

$$\varepsilon_3(\vec{q}) = -2t, \tag{9}$$

see [1]. I also used

$$\cos^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right) = \cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1 \tag{10}$$

.

Interestingly, the top band is completely flat. The other two bands touch when

$$\cos^2(q_x) + \cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1/4 = 0 \tag{11}$$

$$\Rightarrow \cos(q_x) = \frac{1}{2}\left(-\cos\left(\sqrt{3}q_y\right) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1}\right) \tag{12}$$

$$\Rightarrow q_y = \frac{\pi l}{\sqrt{3}},\ l \in \mathbb{N} \tag{13}$$

$$\Rightarrow q_x = (-1)^s\left(\frac{\pi}{2} + \frac{(-1)^l\pi}{6}\right) + 2\pi m,\ m \in \mathbb{N},\ s \in [0,1] \tag{14}$$

A handy solution to this equation is

$$\begin{aligned} q_x &= (-1)^s \frac{2\pi}{3} + m_1\pi \\ q_y &= \frac{\pi}{\sqrt{3}}\left(2m_2 - m_1\right), \end{aligned} \tag{15}$$

with $s \in 0, 1$ and $m_1, m_2 \in \mathbb{N}$. These touching points lie at the corners of the Brillouin zone, they come in two flavours. The parameter s labels the afformentioned flavours while m_1 (m_2) is a translation by the reciprocal lattice vector $\vec{b}_1$ ($\vec{b}_2$). For example

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad s = 0,\ m_1 = 0,\ m_2 = 0 \tag{16}$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad s = 1,\ m_1 = 0,\ m_2 = 0. \tag{17}$$

0.1.4 Filling Factor

The idea is the follwing: The lower two bands contain the same number of states by symmetry. By calculating the number of states of the upper band (should be easy) it shoulb possible to calcualte at what filling factor the low energy physics happen exactly the linear touching points.

0.1.5 Linearization in Reciprocal Space and Projection

Use

$$\cos\left(\left(\vec{K}^{(\prime)}+\vec{p}\right)\cdot\vec{\delta}_{ij}\right)\approx\cos\left(\vec{K}^{(\prime)}\cdot\vec{\delta}_{ij}\right)-\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \tag{18}$$

and

$$\begin{aligned}\vec{K}\cdot\vec{\delta}_1&=\frac{2\pi}{3}\\ \vec{K}\cdot\vec{\delta}_2&=\frac{\pi}{3}\\ \vec{K}\cdot\vec{\delta}_3&=\frac{\pi}{3}\\ \vec{K}'&=-\vec{K}\end{aligned} \tag{19}$$

$$H_{\vec{K}}\approx t\sum_{p\sigma}\begin{pmatrix}a^\dagger_{p\sigma} & b^\dagger_{p\sigma} & c^\dagger_{p\sigma}\end{pmatrix}\begin{pmatrix}0 & 1-\sqrt{3}\vec{p}\cdot\vec{\delta}_2 & 1-\sqrt{3}\vec{p}\cdot\vec{\delta}_3\\ 1-\sqrt{3}\vec{p}\cdot\vec{\delta}_2 & 0 & -1-\sqrt{3}\vec{p}\cdot\vec{\delta}_1\\ 1-\sqrt{3}\vec{p}\cdot\vec{\delta}_3 & -1-\sqrt{3}\vec{p}\cdot\vec{\delta}_1 & 0\end{pmatrix}\begin{pmatrix}a_{q\sigma}\\ b_{q\sigma}\\ c_{q\sigma}\end{pmatrix} \tag{20}$$

Directly at the $\vec{K}$-point the eigenvalues are obviously

$$\varepsilon_1(\vec{K})=t,\quad \varepsilon_2(\vec{K})=t,\quad \varepsilon_3(\vec{K})=-2t, \tag{21}$$

the eigenvector corresponding to the flat band reads at the $\vec{K}$-point:

$$\vec{v}_3(\vec{K})=\frac{1}{\sqrt{3}}\begin{pmatrix}-1\\ 1\\ 1\end{pmatrix}. \tag{22}$$

Since H is hermitian (/symmetric in this case) there exsit a orthonormal basis of eigenvectors. Obviously the subspace orthonormal to $\vec{v}_3(\vec{K})$ corresponds to the two bands that cross. A possible (orthonormal) basis for this subspace is

$$\vec{v}_1(\vec{K})=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\\ 0\end{pmatrix},\quad \vec{v}_2(\vec{K})=\frac{1}{\sqrt{6}}\begin{pmatrix}1\\ -1\\ 2\end{pmatrix}. \tag{23}$$

Projeting H onto this basis yields

$$\begin{aligned}\vec{v}_1^T\cdot H_{\vec{K}}\cdot\vec{v}_1&=t-\sqrt{3}t\vec{\delta}_2\cdot\vec{p} & \vec{v}_1^T\cdot H_{\vec{K}}\cdot\vec{v}_2&=-t\left(\vec{\delta}_1+\vec{\delta}_3\right)\cdot\vec{p}\\ \vec{v}_2^T\cdot H_{\vec{K}}\cdot\vec{v}_1&=-t\left(\vec{\delta}_1+\vec{\delta}_3\right)\cdot\vec{p} & \vec{v}_2^T\cdot H_{\vec{K}}\cdot\vec{v}_2&=t+\sqrt{3}t\vec{\delta}_2\cdot\vec{p}\end{aligned}$$

$$\Leftrightarrow t\sigma_0-\sqrt{3}t\left(\vec{\delta}_2\cdot\vec{p}\right)\sigma_z+t\left(\left(\vec{\delta}_1+\vec{\delta}_3\right)\cdot\vec{p}\right)\sigma_x \tag{24}$$

This already has the correct dispersiob

$$\varepsilon_{1,2}(\vec{K}+\vec{p})\approx\sqrt{3}t|\vec{p}|, \tag{25}$$

but is not yet in the form of the Weyl equation. To transform the equation into the Weyl equation one can rotate the basis inside the degenerate subspace. I take the following approach; define

$$\begin{aligned}\vec{w}_1 &= \alpha\vec{v}_1 + \beta\vec{v}_2 \\ \vec{w}_2 &= \eta\vec{v}_1 + \gamma\vec{v}_2\end{aligned} \tag{26}$$

Othogonality dictates

$$\begin{aligned}|\alpha|^2 + |\beta|^2 &= 1 \\ |\eta|^2 + |\gamma|^2 &= 1 \\ \alpha^*\eta + \beta^*\gamma &= 0.\end{aligned}$$

Maybe make an Ansatz using trig functions and phases? Though I should count free parameters before. Educated guess with 3 free parameters:

$$\begin{aligned}\alpha &= e^{i\theta_1}\sin(\phi) \\ \beta &= e^{i\theta_2}\cos(\phi) \\ \eta &= e^{i\theta_1}\cos(\phi) \\ \gamma &= -e^{i\theta_2}\sin(\phi).\end{aligned}$$

Utilizing $\vec{\delta}_2 + \vec{\delta}_3 = \vec{\delta}_1$ I get:

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = t + \sqrt{3}t\left(\vec{\delta}_2 \cdot \vec{p}\right)\left(1 - 2|\alpha|^2 - \frac{2\mathrm{Re}(\alpha^*\beta)}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3 \cdot \vec{p}\right)\left(-\frac{4\mathrm{Re}(\alpha^*\beta)}{\sqrt{3}}\right) \tag{27}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = 1 + \sqrt{3}t\left(\vec{\delta}_2 \cdot \vec{p}\right)\left(1 - 2|\eta|^2 - \frac{2\mathrm{Re}(\eta^*\gamma)}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3 \cdot \vec{p}\right)\left(-\frac{4\mathrm{Re}(\eta^*\gamma)}{\sqrt{3}}\right) \tag{28}$$

$$\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2 = \sqrt{3}t\left(\vec{\delta}_2 \cdot \vec{p}\right)\left(-2\alpha^*\eta - \frac{\alpha^*\gamma + \beta^*\eta}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3 \cdot \vec{p}\right)\left(-\frac{2(\alpha^*\gamma + \beta^*\eta)}{\sqrt{3}}\right) \tag{29}$$

$$\vec{w}_2^H \cdot H_{\vec{K}} \cdot \vec{w}_1 = \left(\vec{w}_1^H \cdot H_{\vec{K}} \cdot \vec{w}_2\right)^* \tag{30}$$

The strategy is to solve for values for α, β, γ and η such that the following combinations occur in the matrix elements:

$$p_x = \left(\vec{\delta}_2 + \vec{\delta}_3\right) \cdot \vec{p} \tag{31}$$

$$p_y = \frac{1}{\sqrt{3}}\left(\vec{\delta}_2 - \vec{\delta}_3\right) \cdot \vec{p} \tag{32}$$

whil still leading to orthonormal basis vectors. The following parameters are a sensible choice

$$\alpha = -\frac{1}{2}, \quad \beta = \frac{\sqrt{3}}{2}, \quad \eta = \frac{\sqrt{3}}{2}, \quad \gamma = \frac{1}{2}. \tag{33}$$

Thus, the new basis vectors read

$$\vec{w}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-1\\1\end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}2\\1\\1\end{pmatrix} \tag{34}$$

$$\Rightarrow \langle H_{\vec{K}}\rangle_{\vec{w}_1,\vec{w}_2} = t\sigma_0 + \sqrt{3}tp_y\sigma_x + \sqrt{3}tp_x\sigma_z \tag{35}$$

This form was also found in [2], a different one was obtained in [3]. I expect that I would find the latter if I chose different values for α, β, γ and η. I decided to stick with this form to keep the expressions real.

I could eliminate the σ_0 term by introducing chemical potentials but I guess for non-interacting particles it doesn't really matter.

Question: when rotationg the orthonormal basis inside the degnerate subspace there is one free paramters (the rotation angle). On the other hand one can probably also think about the surface of a bloch sphere (the radius is obviously fixed). What is the free parameter in this picture /what is the fixed parameter? It is proably worthwile to explore this at some point, it might reveal the underlying topolgy.

Addition: the counting of free coefficients might not be correct, considering they can be complex. Is the Bloch sphere argument even valid?

0.1.6 Introduction of Strain

I show how to expand expressions prototyical for the k-space Hamiltonian. First I expand in strain:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}}\left(1-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)+i\left(\vec{q}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right), \tag{36}$$

afterwars I expand around $\vec{K}$:

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{q}\cdot\vec{\delta}_{ij}}\left(1+\vec{p}\cdot\vec{\delta}_{ij}\right)\left(1-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)+i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)+i\left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\vec{p}\right). \tag{37}$$

This compares well with the field theory method for the triangular lattice where I did the expansions in reverse order.

Keeping only the relevant terms (relevant to this order at least):

$$t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{p}\cdot\vec{\delta}_{ij}-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)+i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \tag{38}$$

$$\begin{aligned}\Rightarrow t(-\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}}+t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} \approx& 2t\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)-2t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right)\\ &-2\beta t\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\\ &-2t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\end{aligned} \tag{39}$$

I group the linear corrections by order in β to find the two matrices:

$$\delta H_1 = -2t\beta\begin{pmatrix}0 & (u_{xx}+2\sqrt{3}u_{yx}+3u_{yy})/8 & (u_{xx}-2\sqrt{3}u_{yx}+3u_{yy}/8\\ (u_{xx}+2\sqrt{3}u_{yx}+3u_{yy})/8 & 0 & -u_{xx}/2\\ (u_{xx}-2\sqrt{3}u_{yx}+3u_{yy}/8 & -u_{xx}/2 & 0\end{pmatrix} \tag{40}$$

$$\delta H_2 = -2t\begin{pmatrix}0 & \pi(\frac{u_{xx}}{\sqrt{3}}+u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}}-u_{yx})/2\\ \pi(\frac{u_{xx}}{\sqrt{3}}+u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}}\\ \pi(\frac{u_{xx}}{\sqrt{3}}-u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0\end{pmatrix} \tag{41}$$

The projections are

$$\langle\delta H_1\rangle_{\vec{w}_1,\vec{w}_2} = -t\beta\begin{pmatrix}u_{xx} & u_{yx}\\ u_{yx} & u_{yy}\end{pmatrix} \tag{42}$$

$$\langle \delta H_2 \rangle_{\vec{w}_1,\vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} u_{xx} & u_{yx} \\ u_{yx} & -u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} \left(u_{yx}\sigma_y + u_{xx}\sigma_z \right) \tag{43}$$

$\langle \delta H_1 \rangle_{\vec{w}_1,\vec{w}_2}$ can be written as vector potential + scalar potential using:

$$\begin{aligned} u_{xx} &= \frac{1}{2}\left(u_{xx} - u_{yy}\right) + \frac{1}{2}\left(u_{xx} + u_{yy}\right) \\ u_{yy} &= -\frac{1}{2}\left(u_{xx} - u_{yy}\right) + \frac{1}{2}\left(u_{xx} + u_{yy}\right). \end{aligned} \tag{44}$$

$$\Rightarrow \langle H_{\vec{K}} + \delta H_1 + \delta H_2 \rangle_{\vec{w}_1,\vec{w}_2} = (t+\phi)\sigma^0 + \sqrt{3}t\left(p_y + A^y_{\vec{K}}\right)\sigma_x + \sqrt{3}t\left(p_x + A^x_{\vec{K}}\right)\sigma_z \tag{45}$$

$$\phi = \frac{t}{2}\left(u_{xx} + u_{yy}\right) \tag{46}$$

$$A^x_{\vec{K}} = -\frac{t\beta}{2\sqrt{3}}\left(u_{xx} - u_{yy}\right) - \frac{2\pi t}{3}u_{xx} \tag{47}$$

$$A^y_{\vec{K}} = -\frac{t\beta}{\sqrt{3}}u_{yx} - \frac{2\pi t}{3}u_{yx} \tag{48}$$

The β-independent part of $\vec{A}_{\vec{K}}$ can be eliminated using a gauge-transforamtion. This agrees with the results here [3]. It bothers me that only u_{yx} appears but not u_{xy}. Check this

I give a few intermediate results for the derviation of $\vec{A}_{\vec{K}'}$:

The cool kids introduce a cone flavour index and derive the expressioons for both flavours at the same time

$$H_{\vec{K}'} \approx t\sum_{p\sigma}\begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} & c^\dagger_{p\sigma} \end{pmatrix} \begin{pmatrix} 0 & 1+\sqrt{3}\vec{p}\cdot\vec{\delta}_2 & 1+\sqrt{3}\vec{p}\cdot\vec{\delta}_3 \\ 1+\sqrt{3}\vec{p}\cdot\vec{\delta}_2 & 0 & -1+\sqrt{3}\vec{p}\cdot\vec{\delta}_1 \\ 1+\sqrt{3}\vec{p}\cdot\vec{\delta}_3 & -1+\sqrt{3}\vec{p}\cdot\vec{\delta}_1 & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix} \tag{49}$$

(the zeroth order is the same so I make the same Ansatz for the initial basis.)

$$\vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 = t + \sqrt{3}t\vec{\delta}_2 \cdot \vec{p} \qquad \vec{v}_1^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 = t\left(\vec{\delta}_1 + \vec{\delta}_3\right)\cdot\vec{p}$$

$$\vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_1 = t\left(\vec{\delta}_1 + \vec{\delta}_3\right)\cdot\vec{p} \qquad \vec{v}_2^T \cdot H_{\vec{K}'} \cdot \vec{v}_2 = t - \sqrt{3}t\vec{\delta}_2 \cdot \vec{p}$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = t + \sqrt{3}t\left(\vec{\delta}_2\cdot\vec{p}\right)\left(-1 + 2|\alpha|^2 + \frac{2\mathrm{Re}(\alpha^*\beta)}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3\cdot\vec{p}\right)\left(\frac{4\mathrm{Re}(\alpha^*\beta)}{\sqrt{3}}\right) \tag{50}$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = 1 + \sqrt{3}t\left(\vec{\delta}_2\cdot\vec{p}\right)\left(-1 + 2|\eta|^2 + \frac{2\mathrm{Re}(\eta^*\gamma)}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3\cdot\vec{p}\right)\left(\frac{4\mathrm{Re}(\eta^*\gamma)}{\sqrt{3}}\right) \tag{51}$$

$$\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2 = \sqrt{3}t\left(\vec{\delta}_2\cdot\vec{p}\right)\left(2\alpha^*\eta + \frac{\alpha^*\gamma + \beta^*\eta}{\sqrt{3}}\right) + \sqrt{3}t\left(\vec{\delta}_3\cdot\vec{p}\right)\left(\frac{2(\alpha^*\gamma + \beta^*\eta)}{\sqrt{3}}\right) \tag{52}$$

$$\vec{w}_2^H \cdot H_{\vec{K}'} \cdot \vec{w}_1 = \left(\vec{w}_1^H \cdot H_{\vec{K}'} \cdot \vec{w}_2\right)^* \tag{53}$$

$$\Rightarrow \alpha = \frac{\sqrt{3}}{2},\ \beta = \frac{1}{2},\ \eta = \frac{1}{2},\ \gamma = -\frac{\sqrt{3}}{2} \tag{54}$$

$$\Rightarrow \vec{w}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \vec{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \tag{55}$$

These are the same basis vectors as for $\vec{K}$ just in reverse order.

$$\Rightarrow \langle H_{\vec{K}'} \rangle_{\vec{w}_1, \vec{w}_2} = t\sigma_0 + \sqrt{3} t p_y \sigma_x + \sqrt{3} t p_x \sigma_z \tag{56}$$

δH_1 and δH_2 are the same as for $\vec{K}'$ (see 39 and subsitiute $\vec{K} \to \vec{K}' = \vec{K}$). With the reversed order of the basis vectors this gives

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -t\beta \begin{pmatrix} u_{yy} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} \tag{57}$$

$$\langle \delta H_2 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{2\pi t}{\sqrt{3}} \begin{pmatrix} -u_{xx} & u_{yx} \\ u_{yx} & u_{xx} \end{pmatrix} = -\frac{2\pi t}{\sqrt{3}} \left(u_{yx} \sigma_y - u_{xx} \sigma_z \right) \tag{58}$$

$$\phi = \frac{t}{2}(u_{xx} + u_{yy}) \tag{59}$$

$$A^x_{\vec{K}'} = \frac{t\beta}{2\sqrt{3}}(u_{xx} - u_{yy}) + \frac{2\pi t}{3} u_{xx} \tag{60}$$

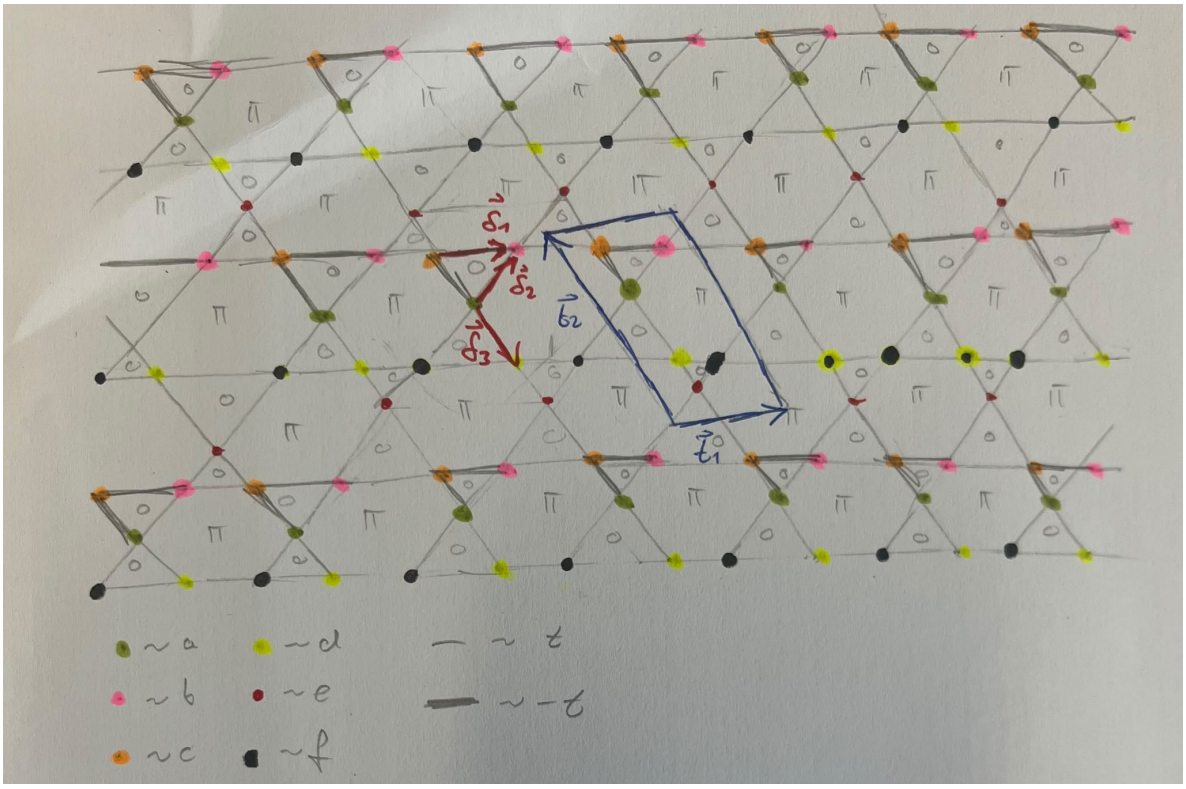
$$A^y_{\vec{K}'} = \frac{t\beta}{\sqrt{3}} u_{yx} + \frac{2\pi t}{3} u_{yx} \tag{61}$$

$$\Rightarrow A_{\vec{K}} = -A_{\vec{K}'} \tag{62}$$

0.2 π-flux State

0.2.1 Real Space

This system is a monoclinic (Is this correct?) Bravais-lattice with a six site unit cell. It emerges from the bare Kagome lattice through unit cell doubling (the stranslational symmetry gets broken by the internal mean field).



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \tag{63}$$

The new primitve translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \tag{64}$$

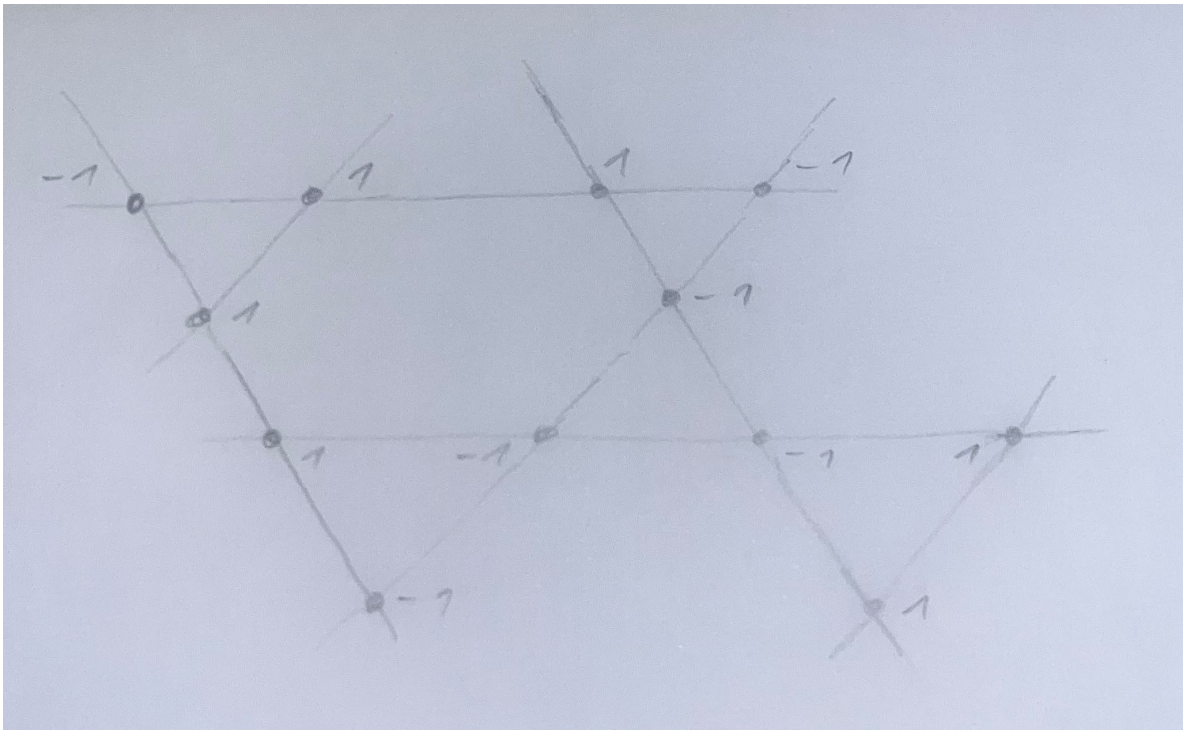
The real space (effective) tight-binding hamiltonian is

$$\begin{aligned} H = t \sum_{i\sigma} \Big(& - a^\dagger_{i\sigma} c_{i\sigma} - c^\dagger_{i\sigma} b_{i\sigma} + b^\dagger_{i\sigma} a_{i\sigma} + a^\dagger_{i\sigma} d_{i\sigma} + d^\dagger_{i\sigma} e_{i\sigma} + e^\dagger_{i\sigma} f_{i\sigma} + f^\dagger_{i\sigma} d_{i\sigma} \\ & + b^\dagger_{i\sigma} c_{i+t_1;\sigma} + b^\dagger_{i\sigma} e_{i+t_1+t_2;\sigma} + f^\dagger_{i\sigma} d_{i+t_1;\sigma} + f^\dagger_{i\sigma} a_{i+t_1;\sigma} + e^\dagger_{i\sigma} c_{i-t_2;\sigma} + \text{h.c.} \Big) \end{aligned} \tag{65}$$

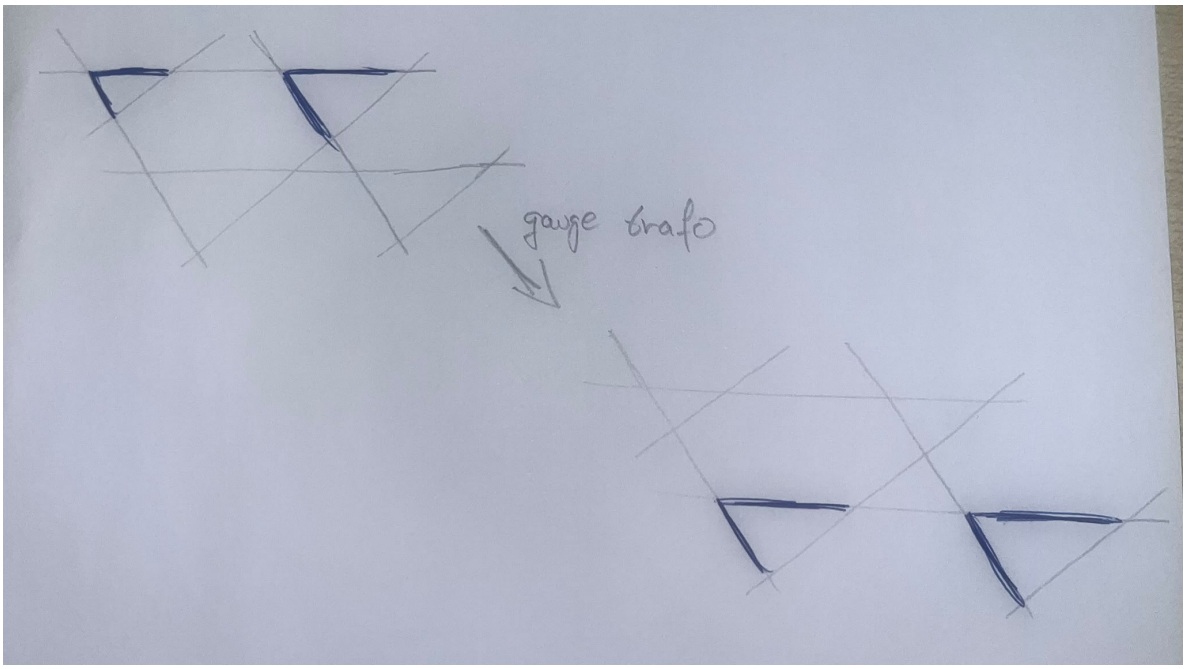
0.2.2 Symmetries

On the level of gauge-invaiant quantuities (that is plaquette fluxes), the system does not break any lattice symmetries, that the bare Kagome lattice has.

The specific gauge choice however looks like it lowers the translational symmetry in one direction ($\vec{t}_2$). Obviously this is a problem of the gauge-choice, the correct symmetries are dictated by the gauge-independent observables. To "repair" this one can supplement lattice symmetry transforamtions with gauge transforamtions, as gauge transformations are always allowed. This whole idea was formally introduced here: [4]. Specifically one can translate the system by $\vec{t}_1/2$ (the symmetry operation of the bare kagome lattice) followed by a gauge transformation. The correct gauge transforamtion in this case is



This has the effect



This then reinstates the correct symmetry. So much for the idea. No for the technical implementation. I define lattice translations

$$T_{\vec{d}} \equiv e^{i\vec{d}\cdot\vec{q}}, \tag{66}$$

,where $\vec{q}$ is the momentum operator (What is the correct momentum operator?), and a gauge transformations

$$G_{\{\theta_i\}} \equiv e^{i\sum_i \theta_i n_i}. \tag{67}$$

What follows is a plethora of questions

- What is the point group of the bare kagome lattice?
- Why are the particle numbers generators of the gauge transformations? (What is the connection to the particle number constraint?)
- Do particle number and momentum commmute? How would I even translate momentum to occupation numbers?
- The description using momentum as quantum numbers must be still correct, no?
- What are the correct quantum numbers considerin the projective symmetry?
- Can one still think about the resciprocal space and which modifications are needed?
- How can one imagine the eigenfunctions of e.g. $G_{\{\theta_i\}} \cdot T_{\vec{t}/2}$?
- The band structure must be gauge-invariant, no?
- How do the bands map under the projective symmetry?

Looking further ahead at strain, can you maybe argue that this forbids pseudo magnetic fields?

0.2.3 Reciprocal Space

The reciprocal lattice vectrors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \tag{68}$$

After a Fourier transform the k-space Hamiltonian reads

$$H = t \sum_{q,\sigma} \Psi^\dagger_{q\sigma} \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i\sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{iq_3} & -2i\sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2\cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2\cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \tag{69}$$

where

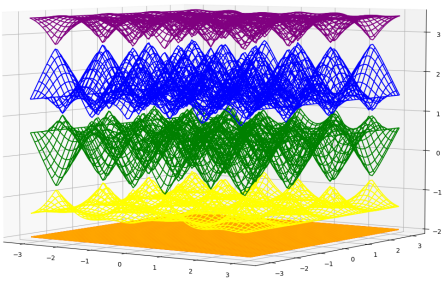
$$\Psi^\dagger_{q\sigma} = \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} & c^\dagger_{q\sigma} & e^\dagger_{q\sigma} & f^\dagger_{q\sigma} & d^\dagger_{q\sigma} \end{pmatrix}. \tag{70}$$

If I write this down somewhere serious, I should just label the sites differently.

I used the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$.

0.2.4 dispersion and band trouching poiunts

The dispersion relation is shwon in the following picture



This plot was generatey by numerically calculating the eigenvalues of H on a (q_x, q_y)-grid. The lowest band (flat band) is doubly degenerate.

The characteristic polynomial of 69 is (this was generate by mathematica)

$$x^6 - 12x^4 - 8x^3 + x^2(30 + 2\Delta_{\vec{q}}) + x(24 + 8\Delta_{\vec{q}}) + 8\Delta_{\vec{q}} - 8 \tag{71}$$

units! with

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{72}$$

I should verify this uisng Cayley-Hamilton. To derive analytical expressions for the bands I first divide out the two flat bands (i.e. polynomial division by $(x+2)^2$). The result is

$$x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}}. \tag{73}$$

To make progress I postulate (judging from the numeriacl plot of the dispersion) that the above polynomial can be written as a product of two quadratic polynomials. Furtheremore the solutions of said qaudratic equations must be centered around t. The p-q forumla yields that the linear coefficents of both quadratic polynomials must therefore be $-2t$.

$$\begin{aligned} \Rightarrow x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}} &= (x^2 - 2x + a) \cdot (x^2 - 2x + b) \\ &= x^4 - 4x^3 + x^2(4 + a + b) - 2x(a + b) + ab. \end{aligned} \tag{74}$$

$$\Rightarrow a + b \overset{!}{=} -4, \quad ab \overset{!}{=} -2 + 2\Delta_{\vec{q}}. \tag{75}$$

These equations are solved by

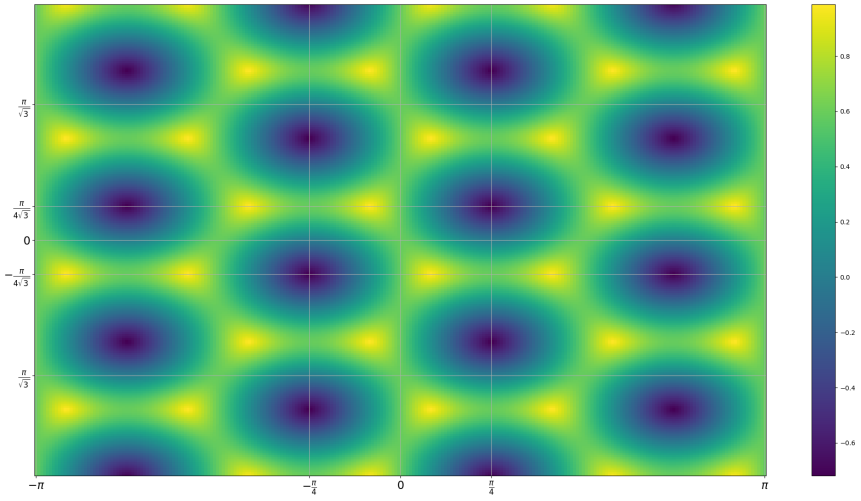
$$\begin{aligned} a_{1,2} &= -2 \pm \sqrt{6 - 2\Delta_{\vec{q}}} \\ b_{1,2} &= -2 \mp \sqrt{6 - 2\Delta_{\vec{q}}}. \end{aligned} \tag{76}$$

Both solution lead to the same result (a and b are exchanged in the second soltuion wrt the first one). This gives the following expressions for the bands

$$
\begin{aligned}
\varepsilon_1(\vec{q}) &= -2t \\
\varepsilon_2(\vec{q}) &= -2t \\
\varepsilon_3(\vec{q}) &= \left(1 - \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_4(\vec{q}) &= \left(1 - \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_5(\vec{q}) &= \left(1 + \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_6(\vec{q}) &= \left(1 + \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t.
\end{aligned} \tag{77}
$$

$$
\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{78}
$$

At half filling the lowest three bands are filled, thus the Fermi-energy is between the yellow and green band.



Shown above is the 4th band (green band). Apparently there are two flavours of touching points with one of each in the first Brillouin zone. They are time revearsel partners. From the numerical data these tow touching points lie at

$$
\vec{K} = \frac{\pi}{4}\begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{K}' = -\frac{\pi}{4}\begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \tag{79}
$$

Now I calculate the various touching points. It is clear from the expressions above that bands 3 and 4 as well as bands 5 and 6 touch iff

$$
\begin{aligned}
& 6 - 2\Delta_{\vec{q}} = 0 \\
\Leftrightarrow\ & -\cos(4q_2) + \cos(4q_3) - \cos(4q_1) = 3 \\
\Leftrightarrow\ & 2\sin(2q_x)\sin\left(2\sqrt{3}q_y\right) - \cos(4q_x) = 3.
\end{aligned} \tag{80}
$$

This can easily be solved considering the ranges of sine and cosine. A possible parametrization of the solution is

$$q_x = (-1)^s \frac{\pi}{4} + m_1\pi \tag{81}$$
$$q_y = (-1)^s \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}} \tag{82}$$

On the other hand, bands 4 and 5 touch when

$$\begin{aligned} &\sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}} = 0 \\ &\Leftrightarrow 2\Delta_{\vec{q}} = -3 \\ &\Leftrightarrow 4\sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) - 2\cos(4q_x) = -3 \end{aligned} \tag{83}$$

Using $\cos(2x) = 1 - 2\sin^2(x)$ I obtain

$$\begin{aligned} &\sin^2(2q_x) + \sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) + \frac{1}{4} = 0 \\ &\Rightarrow \sin(2q_x) = \frac{1}{2}\left(-\sin\Big(2\sqrt{3}q_y\Big) \pm \sqrt{\sin^2(2\sqrt{3}q_y) - 1}\right) \end{aligned} \tag{84}$$

The expression under the square root gives a condition for q_y. It tourns out there are two distinct sets of solutions (meaning there are two distinct sets of equivalent touching points).

$$q_x = (-1)^s \frac{\pi}{12} + m_1\pi \tag{85}$$
$$q_y = (-1)^{s+1} \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}} \tag{86}$$

and

$$q_x = (-1)^s \frac{5\pi}{12} + m_1\pi \tag{87}$$
$$q_y = (-1)^s \frac{\sqrt{3}\pi}{4} + (m_1 + m_2)\frac{\pi}{\sqrt{3}} \tag{88}$$

0.2.5 Linearization in reciproical space and Projection

For sake of the concept I will try to derive the Fermi-velocities of the band-crossings of the 5th band. I will focus on the crossing with the 4th band at

$$\vec{K}''' = \frac{\pi}{12}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \tag{89}$$

The obvious first step is Taylor expansion:

$$\varepsilon_5(\vec{K}''' + \vec{p}) \approx \varepsilon(\vec{K}''') + \left(\frac{\partial \varepsilon_5}{\partial \Delta_{\vec{q}}}\right)\left(\vec{\nabla}_{\vec{q}}\Delta_{\vec{q}}\right)\Big|_{\vec{q}=\vec{K}'''} \cdot \vec{p} + \mathcal{O}(p^2) \tag{90}$$

For now I only do the calculations for $\vec{K}$. Use

$$\vec{K} \cdot \vec{\delta}_1 = \frac{\pi}{4} \tag{91}$$
$$\vec{K} \cdot \vec{\delta}_2 = \frac{\pi}{4} \tag{92}$$
$$\vec{K} \cdot \vec{\delta}_3 = 0 \tag{93}$$

to obtain the linearized Hamiltonian in the vicinity of $\vec{K}$:

$$\begin{pmatrix} 0 & e^{i\frac{\pi}{4}}(1+ip_2) & -1+ip_3 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 \\ e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}i(1+p_1) & e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 \\ -1-ip_3 & -\sqrt{2}i(1+p_1) & 0 & 1-ip_3 & 0 & 0 \\ 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 & 0 & e^{i\frac{\pi}{4}}(1+ip_2) & 1-ip_3 \\ e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}(1-p_1) \\ 1-ip_3 & 0 & 0 & 1+ip_3 & \sqrt{2}(1-p_1) & 0 \end{pmatrix} \tag{94}$$

I should write down a version that is correct at every point in q-space
To keep the size reasonable maybe its a good idea to split this up into zeroth and first order in p. Might also be a good choice for storytelling purposes.

The shorthand means

$$p_i \equiv \vec{\delta}_i \cdot \vec{p},$$

also

$$e^{i\frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}. \tag{95}$$

The eigenvalues for the zeroth order (/directly at the $\vec{K}$-point) are

$$\begin{aligned} &\varepsilon_1(\vec{K}) = -2t && \varepsilon_2(\vec{K}) = -2t && \varepsilon_3(\vec{K}) = (1-\sqrt{3})t \\ &\varepsilon_4(\vec{K}) = (1-\sqrt{3})t && \varepsilon_5(\vec{K}) = (1+\sqrt{3})t && \varepsilon_6(\vec{K}) = (1+\sqrt{3})t \end{aligned} \tag{96}$$

ε_3 and ε_4 are the relevant ones.

I let mathematice calculate orthonormal eigenvectors for H at $\vec{K}$. The relevant ones (that is bands 3 and 4):

$$\vec{v}_3(\vec{K}) = \frac{1}{2\sqrt{6-3\sqrt{3}}}\left(\tfrac{1-\sqrt{3}}{2} + \tfrac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \tfrac{1-\sqrt{3}}{2} + \tfrac{\sqrt{3}-3}{2}i, \quad 0, \quad 1\right)^T \tag{97}$$

$$\vec{v}_4(\vec{K}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}}\left((3-2\sqrt{3}) + (-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3}+(2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3}\right)^T \tag{98}$$

The projections onto those vectors are

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1-\sqrt{3})t - \frac{2t}{\sqrt{3}}p_2 + \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1-\sqrt{3})t + \frac{2t}{\sqrt{3}}p_2 - \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \end{aligned} \tag{99}$$

I transform these to p_x and p_y:

$$
\begin{aligned}
\vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1-\sqrt{3})t + \frac{1-\sqrt{3}}{2}tp_x - \frac{1+\sqrt{3}}{2}tp_y \\
\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1-\sqrt{3})t - \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \\
\vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \\
\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y
\end{aligned}
\tag{100}
$$

The dispersion of this projection is

$$
\begin{aligned}
\varepsilon_{3\vec{K}} &= (1-\sqrt{3})t - \sqrt{2}t\sqrt{p_x^2+p_y^2} \\
\varepsilon_{4\vec{K}} &= (1-\sqrt{3})t + \sqrt{2}t\sqrt{p_x^2+p_y^2}
\end{aligned}
\tag{101}
$$

Most importantly

$$
v_F = \sqrt{2}t \tag{102}
$$

Judging from the pictures of the dispursion I didnt expect it to be so isotropical. Check this.

I follow the usual method defining

$$
\begin{aligned}
\vec{w}_3 &= \alpha\vec{v}_3 + \beta\vec{v}_4 \\
\vec{w}_4 &= \eta\vec{v}_3 + \gamma\vec{v}_4
\end{aligned}
\tag{103}
$$

Orthonormality is garuanted by

$$
|\alpha|^2 + |\beta|^2 = 1 \tag{104}
$$

$$
|\eta|^2 + |\gamma|^2 = 1 \tag{105}
$$

$$
\alpha^*\eta + \beta^*\gamma = 0 \tag{106}
$$

If my counting is correct, there are 8 parameters to begin with (real and imaginary parts), the equations above eliminate 1 and 1 and 2 (the last equation are really 2 equations, one for real and one for imaginary part), leaving behind 4 free parameters. I make the following Ansatz for the leftover parameters:

$$
\begin{aligned}
\alpha &= \cos(\phi/2)e^{i(\chi+\theta/2)}e^{i\kappa} \\
\beta &= \sin(\phi/2)e^{i(\chi+\theta/2)} \\
\eta &= \sin(\phi/2)e^{i(\chi-\theta/2)}e^{i\kappa} \\
\gamma &= -\cos(\phi/2)e^{i(\chi-\theta/2)}
\end{aligned}
\tag{107}
$$

If you want to be fancy about it you can think of what kind of matrix describes the relation between the two bases and then find a suitable parametrization for it. If one thinks of these matrices in terms of groups, maybe there is a suffisticated argument to be made why $U(1)$ appears.. ϕ is a rotation angle, χ a global phase, θ the phase difference between the two states and κ state-internal phase. It is clear that all physical quantities can not depend on χ. At this point I might as well choose $\chi = 0$.

The following relations are impotant to derive the matrix elements in the rotated basis:

$$
\begin{aligned}
(2|\alpha|^2 - 1) &= \cos(\phi) \\
2\mathrm{Re}(\alpha^*\beta) &= \sin(\phi)\cos(\kappa) \\
\alpha^*\eta - \beta^*\gamma &= \sin(\phi)\cos(\theta) - i\sin(\phi)\sin(\theta) \\
\alpha^*\gamma + \beta^*\eta &= \sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta) \\
&\quad + i\big(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\big)
\end{aligned}
$$

Using these, in general (while neglecting the σ_0-term),

$$
\vec{w}_1^H \cdot H \cdot \vec{w}_1 \sim \big(\vec{v}_1^H \cdot H \cdot \vec{v}_1\big)\cos(\phi) + \big(\vec{v}_1^H \cdot H \cdot \vec{v}_2\big)\sin(\phi)\cos(\kappa) \tag{108}
$$

$$
\begin{aligned}
\vec{w}_1^H \cdot H \cdot \vec{w}_2 = &\big(\vec{v}_1^H \cdot H \cdot \vec{v}_1\big)\sin(\phi)\cos(\theta) + \big(\vec{v}_1^H \cdot H \cdot \vec{v}_2\big)\left(\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right) \\
&- i\left[\big(\vec{v}_1^H \cdot H \cdot \vec{v}_1\big)\sin(\phi)\sin(\theta) - \big(\vec{v}_1^H \cdot H \cdot \vec{v}_2\big)\left(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right)\right]
\end{aligned} \tag{109}
$$

I find for the projection in the transformed basis (I again drop the σ_0 part as it transforms trivially):

$$
\begin{aligned}
\frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} \sim\ & \frac{1}{\sqrt{8}}\Big((1-\sqrt{3})\cos(\phi) - (1+\sqrt{3})\sin(\phi)\cos(\kappa)\Big)\,p_x \\
&+ \frac{1}{\sqrt{8}}\Big(-(1+\sqrt{3})\cos(\phi) - (1-\sqrt{3})\sin(\phi)\cos(\kappa)\Big)\,p_y
\end{aligned}
$$

$$
\begin{aligned}
\frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} \sim\ & \frac{1}{\sqrt{8}}\Big(-(1-\sqrt{3})\cos(\phi) + (1+\sqrt{3})\sin(\phi)\cos(\kappa)\Big)\,p_x \\
&+ \frac{1}{\sqrt{8}}\Big((1+\sqrt{3})\cos(\phi) + (1-\sqrt{3})\sin(\phi)\cos(\kappa)\Big)\,p_y
\end{aligned}
$$

$$
\begin{aligned}
\frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} = & \frac{1}{\sqrt{8}}\Big((1-\sqrt{3})\sin(\phi)\cos(\theta) - (1+\sqrt{3})\left[\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right]\Big)\,p_x \\
&+ \frac{1}{\sqrt{8}}\Big(-(1+\sqrt{3})\sin(\phi)\cos(\theta) - (1-\sqrt{3})\left[\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right]\Big)\,p_y \\
&- \frac{i}{\sqrt{8}}\Big((1-\sqrt{3})\sin(\phi)\sin(\theta) + (1+\sqrt{3})\left[\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right]\Big)\,p_x \\
&- \frac{i}{\sqrt{8}}\Big(-(1+\sqrt{3})\sin(\phi)\sin(\theta) + (1-\sqrt{3})\left[\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right]\Big)\,p_y
\end{aligned}
$$

$$
\begin{aligned}
\frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} = & \frac{1}{\sqrt{8}}\Big((1-\sqrt{3})\sin(\phi)\cos(\theta) - (1+\sqrt{3})\left[\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right]\Big)\,p_x \\
&+ \frac{1}{\sqrt{8}}\Big(-(1+\sqrt{3})\sin(\phi)\cos(\theta) - (1-\sqrt{3})\left[\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right]\Big)\,p_y \\
&+ \frac{i}{\sqrt{8}}\Big((1-\sqrt{3})\sin(\phi)\sin(\theta) + (1+\sqrt{3})\left[\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right]\Big)\,p_x \\
&+ \frac{i}{\sqrt{8}}\Big(-(1+\sqrt{3})\sin(\phi)\sin(\theta) + (1-\sqrt{3})\left[\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right]\Big)\,p_y
\end{aligned}
$$

From these equations the $\sigma_{0,x,y,z}$-components can be read of. I decided to solve for an equation of the form $H = \varepsilon_0\sigma_0 + v_F(p_x\sigma_x + p_y\sigma_z)$. The corresponding angles are

$$\begin{aligned}
\cos(\phi) &= -\frac{1+\sqrt{3}}{\sqrt{8}} \quad \Rightarrow \quad \phi = \frac{11}{12}\pi \\
\cos(\kappa) &= -\frac{1-\sqrt{3}}{\sqrt{4-2\sqrt{3}}} = 1 \quad \Rightarrow \quad \kappa = 0 \\
\cos(\theta) &= \frac{1-\sqrt{3}}{\sqrt{4-2\sqrt{3}}} = -1 \quad \Rightarrow \quad \theta = \pi.
\end{aligned} \tag{110}$$

$\sqrt{4-2\sqrt{3}} = \sqrt{3}-1$:D. First I had lhs=$1-\sqrt{3}$, and it took me around 3 hours to find the mistake.

Another solution corresponding to the form $H = \varepsilon_0\sigma_0 + v_F(p_x\sigma_x + p_y\sigma_y)$ is

$$\begin{aligned}
\phi &= -\frac{\pi}{2} \\
\kappa &= -\frac{\pi}{2} \\
\theta &= \frac{5}{12}\pi
\end{aligned} \tag{111}$$

I will continue to work with this solution from now on. Typically I will employ the first solution as a sanity check.

0.2.6 Introduction of strain

The relevant equations are

$$t(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1 + i\vec{p}\cdot\vec{\delta}_{ij} - \beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) + i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \tag{112}$$

and

$$\begin{aligned}
t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(-\vec{\delta}_{ij})e^{-i\vec{K}\cdot\vec{\delta}_{ij}} \approx & 2t\cos\left(\vec{q}\cdot\vec{\delta}_{ij}\right) - 2t\sin\left(\vec{q}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\
& -2\beta t\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\
& -2t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)
\end{aligned} \tag{113}$$

and

$$\begin{aligned}
t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} - t(-\vec{\delta}_{ij})e^{-i\vec{q}\cdot\vec{\delta}_{ij}} \approx & 2it\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right) + 2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\
& -2i\beta t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\
& +2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)
\end{aligned} \tag{114}$$

I split the everything into two parts:

$$\begin{aligned}
\langle\delta H_1\rangle_{\vec{w}_1,\vec{w}_2} &= -\frac{\beta}{2}(1-\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \\
\langle\delta H_2\rangle_{\vec{w}_1,\vec{w}_2} &= \sqrt{2}\frac{\pi}{4}\left(u_{xx} + \frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy} + \frac{u_{yy}}{\sqrt{3}}\right)\sigma_y
\end{aligned} \tag{115}$$

The second term is a gradient field and can be gauged away. This means that to this order there is no effective vector potential but only an effective scalar potential.

Doing the same calculation for K' basically means transforming the Hamiltonian

$$H \mapsto \tilde{H} \equiv THT, \tag{116}$$

with T the time revearsel. Thus, the eigenvectors of $\tilde{H}$ are $T\vec{v}_1$ and $T\vec{v}_2$ (since T is its own inveres). In this case time revearsel (which can be interpreted as either $\vec{K} \mapsto -\vec{K}$ or $\vec{q} \mapsto -\vec{q}$) is equivalent to complex conjugation. Due to the antilinearity of time revearsel, the matrix elements are conjugated. Thus the parameters have to be adjusted.

0.2.7 crossing 5-6

Since bands 5 and 6 cross at the same point as bands 3 and 4, the expansions of the Hamiltonian are still valid. The eigenvectors corresponding to bands 5 and 6 at the K-point are

$$\vec{v}_5(\vec{K}) = \frac{1}{\sqrt{\sqrt{6}+3\sqrt{2}}}\left(-\tfrac{1}{\sqrt{2}} - i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad -2-\sqrt{3} \quad i\sqrt{2}(1+\sqrt{3}) \quad -\tfrac{1}{\sqrt{2}} + i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad 1 \quad 0\right) \tag{117}$$

$$\vec{v}_6(\vec{K}) = \frac{1}{4\sqrt{3}}\left(1+\sqrt{3}+i(1-\sqrt{3}) \quad \sqrt{2}+\sqrt{6} \quad 0 \quad 1+\sqrt{3}-i(1-\sqrt{3}) \quad \sqrt{2}+\sqrt{6} \quad 4\right) \tag{118}$$

The projections otno $\vec{v}_5(\vec{K})$ and $\vec{v}_6(\vec{K})$ are

$$\begin{aligned}
\vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_5 &= (1+\sqrt{3})t + \frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \\
\vec{v}_6^H \cdot H_{\vec{K}} \cdot \vec{v}_6 &= (1+\sqrt{3})t - \frac{1+\sqrt{3}}{2}tp_x + \frac{1-\sqrt{3}}{2}tp_y \\
\vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_6 &= \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \\
\vec{v}_6^H \cdot H_{\vec{K}} \cdot \vec{v}_5 &= \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y
\end{aligned} \tag{119}$$

$$\Rightarrow \langle H_{\vec{K}} \rangle_{\vec{v}_5,\vec{v}_6} = t(1+\sqrt{3})\sigma_0 + \frac{t}{2}\left((1+\sqrt{3})p_x - (1-\sqrt{3})p_y\right)\sigma_z + \frac{t}{2}\left((1-\sqrt{3})p_x + (1+\sqrt{3})p_y\right)\sigma_x \tag{120}$$

Thus, the dispersions are

$$\varepsilon_{5\vec{K}}(\vec{p}) = (1+\sqrt{3})t - \sqrt{2}t\sqrt{p_x^2+p_y^2} \tag{121}$$

$$\varepsilon_{6\vec{K}}(\vec{p}) = (1+\sqrt{3})t + \sqrt{2}t\sqrt{p_x^2+p_y^2} \tag{122}$$

Again the Fermi velocity is $v_F = \sqrt{2}t$. I use the usual method and refine it a bit. Consider the coefficients

$$H_{11}\cos(\phi) + H_{12}\sin(\phi)\cos(\kappa) \qquad (\sigma_z) \tag{123}$$

$$H_{11}\sin(\phi)\cos(\theta) + H_{12}\left(\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right) \qquad (\sigma_x) \tag{124}$$

$$H_{11}\sin(\phi)\sin(\theta) - H_{12}\left(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right) \qquad (\sigma_y) \tag{125}$$

I used the shorthand $H_{ii} \equiv \vec{v}_i^T \cdot H_{\vec{K}} \cdot \vec{v}_i$. Going forward I split these coefficients into p_x and p_y parts, which I should call H_{ii}^x and H_{ii}^y, such that $H_{ii} = H_{ii}^x p_x + H_{ii}^y p_y$.

Solving for an equation of the form $H = \varepsilon_0\sigma_o + p_x\sigma_x + p_y\sigma_y$, eq 123 directly gives

$$\cos(\kappa) = 0$$
$$\cos(\phi) = 0,$$

assuming that $H_{11}^{x/y} \neq 0$ and $H_{12}^{x/y} \neq 0$. This is not enough to fix κ and ϕ yet as both $\pi/2$ and $-\pi/2$ are valid solutions. (I lack a rigorous arguent here). Equations 124 and 125 reveal

$$\tan(\theta) = -\frac{H_{12}^x}{H_{12}^y}. \tag{126}$$

In this case that means

$$\begin{aligned} \theta &= \arctan\left(-\frac{1-\sqrt{3}}{1+\sqrt{3}}\right) = \frac{\pi}{12} \\ \phi &= \pi/2 \\ \kappa &= -\pi/2 \end{aligned} \tag{127}$$

This resulting strain is

$$\langle \delta H_1 \rangle_{\vec{w}_5,\vec{w}_6} = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \tag{128}$$
$$\langle \delta H_2 \rangle_{\vec{w}_5,\vec{w}_6} = \sqrt{2}\frac{\pi}{4}\left(u_{xx} + \frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy} + \frac{u_{yy}}{\sqrt{3}}\right)\sigma_y \tag{129}$$

Apparently the scalar potential scales with the energy?

0.2.8 crossing 4-5

I begin with the degneracy point at

$$\vec{K}'' = \frac{\pi}{12}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}, \tag{130}$$

It should be easy to verfy the following calculations by repeating them for $-\vec{K}''$.

I pick the orthonormal basis vectors

$$\vec{v}_4(\vec{K}'') = \frac{1}{\sqrt{6}\sqrt{5\sqrt{3}+9}}\begin{pmatrix} \frac{1+\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2+\sqrt{3}) & 2+\sqrt{3} & i\sqrt{2}(2+\sqrt{3}) & \frac{1+\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2+\sqrt{3}) & 1 & 0 \end{pmatrix} \tag{131}$$

$$\vec{v}_5(\vec{K}'') = \frac{\sqrt{2}}{\sqrt{3\sqrt{3}(1+\sqrt{3})}}\begin{pmatrix} -\frac{1}{2} & -\frac{3}{2\sqrt{2}}(1+\sqrt{3}) & i\frac{1+\sqrt{3}}{2} & -\frac{1}{2} & \frac{1+\sqrt{3}}{2\sqrt{2}} & 1 \end{pmatrix} \tag{132}$$

The projections are

$$\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 = -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \tag{133}$$
$$\vec{v}_5^H \cdot H_{\vec{K}} \cdot \vec{v}_5 = \frac{1+\sqrt{3}}{2}tp_x + \frac{1-\sqrt{3}}{2}tp_y \tag{134}$$
$$\vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_5 = -\frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \tag{135}$$

$$\Rightarrow v_F = \sqrt{2}t \tag{136}$$

This yields the angles

$$\begin{aligned}\theta &= \arctan\left(\frac{1-\sqrt{3}}{1+\sqrt{3}}\right) = -\pi/12\\ \phi &= -\pi/2\\ \kappa &= -\pi/2\end{aligned} \tag{137}$$

$$\Rightarrow \langle H_{\vec{K}}\rangle_{\vec{w}_1,\vec{w}_2} = \sqrt{2}tp_x\sigma_x + \sqrt{2}tp_y\sigma_y \tag{138}$$

The resulting lowest order terms are

$$\langle \delta H_1\rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_y \tag{139}$$

$$\langle \delta H_2\rangle_{\vec{w}_4,\vec{w}_5} = \sqrt{2}\frac{\pi}{12}(u_{xx}-\sqrt{3}u_{yx})\sigma_x + \sqrt{2}\frac{\pi}{12}(u_{xy}-\sqrt{3}u_{yy})\sigma_y \tag{140}$$

Since the second term can be written as a gradient field only the first term contributes to a pseudo-magentic field. I find

$$\phi(\vec{K}'') = -\frac{\beta}{2}(u_{xx}+u_{yy}) \tag{141}$$

$$\vec{A}(\vec{K}'') = \frac{\beta}{2\sqrt{3}}\begin{pmatrix}u_{yy}-u_{xx}\\ u_{xy}+u_{yx}\end{pmatrix}. \tag{142}$$

$$\Rightarrow B(\vec{K}'') = \frac{\beta}{2\sqrt{3}}\left(2\partial_x\partial_y U_x + \left[\partial_x^2-\partial_y^2\right]U_y\right) \tag{143}$$

For the purpose of verification I do the same calculation for an effective theory of the form $H = \varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_z$. 125 directly gives

$$\sin(\theta) = 0 \tag{144}$$

$$\sin(\kappa) = 0. \tag{145}$$

Furthermore, from eqs 124 and 125:

$$\tan(\phi) = \sqrt{-\frac{H_{11}^x H_{12}^y}{H_{11}^y H_{12}^x}}. \tag{146}$$

This is not yet enough to fix the sign of ϕ. I'll have to return to this to fix θ and κ, the problem is in structure identical to the other effective theory above.

The right choice is thus

$$\phi = \arctan\left(\sqrt{7+4\sqrt{3}}\right) = \frac{5\pi}{12} \tag{147}$$

$$\kappa = 0 \tag{148}$$

$$\theta = \pi \tag{149}$$

The projections onto those vectors are

$$\langle \delta H_1 \rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{yy}+u_{xx})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_z \tag{150}$$

$$\langle \delta H_2 \rangle_{\vec{w}_4,\vec{w}_5} = ? \tag{151}$$

Now I repeat the same for the other set of degeneracy points. Among those I pick

$$\vec{K}''' = \frac{\pi}{12}\begin{pmatrix} 5 \\ 3\sqrt{3} \end{pmatrix} \tag{152}$$

I choose the vectors

$$v_4(\vec{K}''') = \frac{1}{\sqrt{6}\sqrt{9-5\sqrt{3}}}\begin{pmatrix} \frac{1-\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2-\sqrt{3}) & 2-\sqrt{3} & i\sqrt{2}(\sqrt{3}-2) & \frac{1-\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2-\sqrt{3}) & 1 & 0 \end{pmatrix} \tag{153}$$

$$v_5(\vec{K}''') = \frac{\sqrt{2}}{\sqrt{3}\sqrt{3-\sqrt{3}}}\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2\sqrt{2}}(1-\sqrt{3}) & i\frac{1-\sqrt{3}}{2} & \frac{1}{2} & -\frac{1}{2\sqrt{2}}(1-\sqrt{3}) & 1 \end{pmatrix} \tag{154}$$

The projection coefficents are

$$\begin{aligned} H^x_{11} &= -\frac{1-\sqrt{3}}{2}t \\ H^y_{11} &= \frac{1+\sqrt{3}}{2}t \\ H^x_{12} &= -\frac{1+\sqrt{3}}{2}t \\ H^y_{12} &= -\frac{1-\sqrt{3}}{2}t. \end{aligned} \tag{155}$$

For a theory of the Form $H = \varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ the correct parameters are

$$\begin{aligned} \theta &= \arctan\left(2+\sqrt{3}\right) = \frac{5\pi}{12} \\ \kappa &= -\pi/2 \\ \phi &= \pi/2 \end{aligned} \tag{156}$$

The projections are

$$\langle \delta H_1 \rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xx}-u_{yy})\sigma_x - \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_y \tag{157}$$

$$\langle \delta H_2 \rangle_{\vec{w}_4,\vec{w}_5} = \sqrt{2}\frac{\pi}{12}\left(5u_{xx}+3\sqrt{3}u_{yx}\right)\sigma_x + \sqrt{2}\frac{\pi}{12}(5u_{xy}+3\sqrt{3}u_{yy})\sigma_y \tag{158}$$

$$\phi(\vec{K}''') = -\frac{\beta}{2}(u_{xx}+u_{yy}) \tag{159}$$

$$\vec{A}(\vec{K}''') = \frac{\beta}{2\sqrt{3}}\begin{pmatrix} u_{xx}-u_{yy} \\ -(u_{xy}+u_{yx}) \end{pmatrix} \tag{160}$$

It turns out $\vec{A}(\vec{K}'') = -\vec{A}(\vec{K}''')$. The resulting magnetic field is

$$B(\vec{K}'') = -\frac{\beta}{2\sqrt{3}}\left(2\partial_x\partial_y U_x + \left[\partial_x^2 - \partial_y^2\right]U_y\right). \tag{161}$$

0.3 projection method

Here I collect the method. Asssume a k-space Hamiltonian $H(\vec{q})$ and a degeneracy point $\vec{K}$. Furthermore, I define

$$H(\vec{K}+\vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}}H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{162}$$

The next ingredient is a orthonormal basis of the degenerate subspace $\{\vec{v}_1, \vec{v}_2\}$. I also need the matrix elements of $H^1_{\vec{K}}$ wrt to this basis:

$$\begin{aligned}
\vec{v}_1^H \cdot H^1_{\vec{K}} \cdot \vec{v}_1 &\equiv M_{11} = M^x_{11}p_x + M^y_{11}p_y \\
\vec{v}_1^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &\equiv M_{12} = M^x_{12}p_y + M^y_{12}p_y \\
\vec{v}_2^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &= -M_{11} \\
\vec{v}_2^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &= (M_{12})^*
\end{aligned} \tag{163}$$

The last two identities are motivated by the effect that $(M)_{ij} = M_{ij}$ must be a Hermitian matrix with no absolute part ($\propto \sigma_0$). To obtain an arbitrary basis of the degnerate subspace I do a $SU(2)$-rotation ($SU(2)$ automatically enforces the orthonormality of the new basis):

$$\begin{pmatrix}\vec{w}_1\\ \vec{w}_2\end{pmatrix} = \begin{pmatrix}e^{i\chi} & 0\\ 0 & e^{i\chi}\end{pmatrix}\begin{pmatrix}\cos(\phi/2)e^{i(\theta/2+\kappa)} & \sin(\phi/2)e^{i\theta/2}\\ \sin(\phi/2)e^{i(-\theta/2+\kappa)} & -\cos(\phi/2)e^{-i\theta/2}\end{pmatrix}\begin{pmatrix}\vec{v}_1\\ \vec{v}_2\end{pmatrix} \tag{164}$$

Parameter ranges?Optimally I would find a Group-theoretical justification of this representaion. χ is a global phase, which I choose to be 0, justified by the $U(1)$-gauge symmetry of quantum mechanics. This leaves three parameters. Now I can write down the projection of $H^1_{\vec{K}}$ onto $\{\vec{w}_1, \vec{w}_2\}$. I write the resulting hermitian 2x2 matrix in the Pauli-matrix basis (again there is no σ_0-part):

$$\begin{aligned}
\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} = \quad & \sigma_x\left[M_{11}\sin(\phi)\cos(\theta) + M_{12}\left(\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right)\right] && (165)\\
& +\sigma_y\left[M_{11}\sin(\phi)\sin(\theta) - M_{12}\left(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right)\right] && (166)\\
& +\sigma_z\left[M_{11}\cos(\phi) + M_{12}\sin(\phi)\cos(\kappa)\right] && (167)
\end{aligned}$$

One might complain that the equations above dont reference the Fermi velocity. However, as long as we only make statements using ratios of M-components we still obtain the desired result. Now, (ϕ, θ, κ) can be choosen to i.e. give a projected Hamiltonian of the form

$$\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F\left(p_x\sigma_x + p_y\sigma_y\right). \tag{168}$$

A partial solution is

$$\begin{aligned}
\cos(\phi) &= 0 \\
\cos(\kappa) &= 0 \\
\tan(\theta) &= -\frac{M^x_{12}}{M^y_{12}}.
\end{aligned} \tag{169}$$

The above is not sufficient to completely fix ϕ and κ as both $\pi/2$ and $-\pi/2$ are still correct solutions, yet only one combination gives the correct result. However, the partial solutions ensures that the σ_z component vanishes in every case. At the moment I just have to try all 4 possibilities until I found a systematic solution. I expect that the answer lies in the signs of the M_{ij}.

One could (for example) also pick the form

$$\langle H^1_{\vec{K}} \rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F \big(p_x \sigma_x + p_y \sigma_z\big). \tag{170}$$

A partial solution for this is

$$\begin{aligned} \tan(\phi) &= \sqrt{-\frac{M^x_{11} M^y_{12}}{M^y_{11} M^x_{12}}} \\ \sin(\kappa) &= 0 \\ \sin(\theta) &= 0. \end{aligned} \tag{171}$$

Again similar problem as above, κ and θ are not completely fixed yet. In what follows I will usually apply the second form as a consistency check.

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